Bridge Infrastructure Asset Management System: Comparative Computational Machine Learning Approach for Evaluating and Predicting Deck Deterioration Conditions

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Abstract: Bridge infrastructure asset management system is a prevailing approach toward having an effective and efficient procedure for monitoring bridges through their different development phases including construction, operation, and maintenance. Damage to any structural component of a bridge will negatively affect its safety, integrity, and longevity. Bridge decks are more susceptible to severe deterioration because they are exposed to harsh conditions including heavy traffic, varying temperatures, road salts, and abrasive forces. The ability to forecast the conditions of bridges in an accurate way has been a great challenge to transportation agencies. Many previous research studies highlighted the need to have a data-driven approach in predicting and evaluating the deterioration conditions of bridges. As such, this paper develops a computational data-driven asset management system to evaluate and predict bridge deck deterioration conditions. A multistep interdependent research methodology was utilized. First, the best set of variables affecting the conditions of bridge decks was identified. Second, two computational machine learning models were developed for the prediction of deck conditions using artificial neural networks (ANNs) and *k*-nearest neighbors (KNNs). Third, a comparison between the developed models is conducted to select the ultimate model with the highest accuracy. The result is a framework that is able to evaluate and predict deck conditions with a prediction accuracy of 91.44%. While this research is applied to bridges in Missouri, the technique can be used on any similarly available data set nationwide. This study adds to the body of knowledge by devising a computational data-driven framework that is valuable for transportation agencies as it allows them to evaluate and predict deck conditions with high accuracy. Consequently, this will help in ensuring proper and effective distribution of funds allocated for the maintenance, rehabilitation, and repair of bridges. Ultimately, this will result in minimizing efforts, time, and costs associated with site inspection of bridge decks. **DOI: 10.1061/(ASCE)IS.1943-555X.0000572.**

Introduction

Bridges are one of the most important infrastructure systems. Bridges are composed of many interdependent structural elements; any damage to these individual components dramatically affects the structural safety and integrity of bridges (Tung et al. 2002). To this end, periodic inspections and evaluations of the conditions of the different structural parts for bridges are crucial to reflect their overall deterioration condition. In addition, continuous planning of maintenance, repair, and rehabilitation (MR&R) for bridges is needed to ensure public safety. The assessment of conditions for bridges in the US is still largely performed by visual and site/field inspection (Graybeal et al. 2002). According to the National Bridge Inspections Standards Regulation (NBIS), the inspection frequency of bridges should not exceed 24 months (FHWA 2018b). The evaluation of the structural elements of a bridge is based on the personal experiences and subjective assessments of the inspectors (Phares et al. 2004) while still being dictated by the guidelines specified by the FHWA.

Compared to other structural elements, bridge decks are more susceptible to severe deterioration because they are exposed to harsh conditions such as heavy traffic, varying temperatures, road salts, and abrasive forces (Alberta Transportation 2003). States departments of transportation and the USDOT have been facing escalating challenges for decades due to the deterioration of bridge decks (Buckler et al. 2000). In addition, the current and future conditions of bridge decks are considered to affect the amount of future bridge funding needs (Abed-Al-Rahim and Johnston 1995). Moreover, the deterioration of bridge decks could adversely affect the serviceability of bridges (Scott et al. 2003; Kirkpatrick et al. 2002). As such, monitoring the conditions of bridge decks is very important in preventing any damage that could occur due to inadequate assessments and evaluations.

The ability to forecast the conditions of bridges in an accurate way has been a great challenge to transportation agencies. Liu and El-Gohary (2017) stressed this by specifying that: "a large amount of data about bridge conditions and maintenance actions are buried in textual bridge inspection reports and are not utilized [because] bridge owners are unable to convert the 'mountain' of inspection data in bridge inspection reports into effective maintenance decisions." Further, since bridge inspections require huge efforts, much time, high costs, and substantial danger (Gillins et al. 2016),

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transportation agencies directed their interests and efforts to develop data-driven tools that could be adopted quickly and easily as first estimates while still providing accurate forecasts for the conditions of bridge decks. To this end, this paper attempts to help transportation agencies by developing a computational data-driven decision-support tool based on machine learning algorithms for the assessment and prediction of the condition of bridge decks.

Goal and Objectives

The goal of this paper is to develop a bridge management system (BMS) that can evaluate and predict bridge deck deterioration conditions. The associated objectives include: (1) identifying the best subset of factors that affect the classification of the conditions of bridge decks; (2) investigating the performance of two machine learning techniques: artificial neural networks (ANNs) and *k*-nearest neighbors (KNNs); and (3) optimizing the accuracy of the developed classification machine learning algorithms by finding the best set of hyperparameters that lead to the highest performance. While this research is applied to bridges in Missouri, the technique is scalable and malleable enough to be used on any similarly available data set nationwide. This study should be valuable for transportation agencies as it allows them to evaluate and predict deck conditions with high accuracy. Consequently, this will help in ensuring proper and more effective distribution of funds allocated for the MR&R of bridges. Ultimately, this will result in minimizing efforts, time, and costs associated with site inspection of bridge decks.

Background Information

Review of the Technical Literature and Comparison between Different Options

This section aims to provide a review of the technical literature on the deterioration of bridges and to compare various conventional alternatives because previous studies attempted to evaluate the conditions of bridges using different methods. This is important since new research should build on previous efforts to convey prospective findings that add to the body of knowledge (El-adaway et al. 2019; Assaad and Abdul-Malak 2020b). In relation to the technical literature, Manafpour et al. (2018) used stochastic analysis and time-based modeling to evaluate the deterioration of concrete bridge decks and tried to relate the deterioration rates to different aspects including structural system attributes, types of rebar protection, continuous/simply supported spans, overall bridge deck length, types of overlay, average daily traffic, route type, and environmental conditions. Liang and Parlikad (2016) developed a framework for concrete decks by modeling the chloride-induced deterioration as an induced accelerated deterioration through a continuous-time approach with multiple-dependent deterioration paths. Rhee et al. (2018) investigated the deterioration characteristics of concrete bridge decks with asphalt concrete by delineating the relationship between the deterioration status of the decks and the asphalt overlay, waterproof layer, cover thickness of rebars, corrosion of rebar, and chloride contents. Hu et al. (2016b) developed a framework for estimating bridge deck chloride-induced degradation from local modeling to global asset assessment using a Monte Carlo simulation and a deterministic mechanics-based model. Lee et al. (2015) developed a linear regression model to analyze the deterioration of highway bridge deck slab deterioration and determined correlations based on the number of snowy days, the amount of snowfall, the number of freeze–thaw days, and average winter temperature, among others. Lou et al. (2016) studied the effect of overweight trucks on bridge deck deterioration based on weigh-in motion data by deriving the deterioration of decks over time and the expected service lives of decks on different highways. Although previous research studies have developed frameworks for the assessment of the deteriorations of bridges, these efforts are still considered to be limited in terms of the modeling techniques used; especially, given the fact that the data is easily available (Contreras-Nieto et al. 2018).

Furthermore, the construction industry is very complex (Assaad and El-adaway 2020a; Assaad and Abdul-Malak 2020a) with infrastructures being one of the most important and complicated construction sectors. That said, bridge inspections are not simple and are performed to identify different defects in the decks including: cracking, scaling, spalling, leaching, chloride contamination, potholing, delamination, full or partial depth failures, broken welds, broken grids, section loss, growth of filled grids from corrosion splitting, crushing, fastener failure, and deterioration from rot (Weseman 1995). The identification of such defects in bridge decks could be performed by the methods known as conventional or state-of-the-art alternatives that appear in the literature. The first conventional method is visually through a technical field visit by authorized engineers. For field inspections, all bridge decks should receive a close visual inspection; however, physical examinations must also be used for suspect areas, and they could include: sounding, probing, drilling, core sampling, electrical testing, etc. (Rossow 2019). Nevertheless, since the visual/field inspection requires a huge effort, substantial human subject matter expertise, time, and substantial danger (Gillins et al. 2016), many US states started to rely on alternatives. The second conventional method is the use of nondestructive technological methods that include different techniques such as: ground penetrating radar, infrared thermography, and high-resolution imaging (Penetradar 2019). Ground penetrating radar is a nonintrusive mean of determining the condition of asphalt covered and exposed concrete bridge decks using multiple noncontact antennas. The antennas' scans are processed and combined to produce a plan view mapping of the bridge deck showing the size and shape of the delamination and deteriorated area through a graphical output format. On the other hand, infrared thermography uses a thermographic camera and employs leading-edge computer-based video data acquisition and image processing software and hardware to detect delaminations, deteriorated deck areas, and debonded overlays. Similarly, high-resolution imaging is based on a video mapping framework and a specialized software that transforms the high-resolution video into plan-view images of bridge decks to detect deteriorations and voids beneath pavements. Although the nondestructive technological methods address some of the shortcomings of the visual/field inspection; they do not completely overcome all difficulties. The reason behind this is that their effectiveness is site specific, varies greatly from place-to-place (CNI Locates 2019), and depends on the flatness and evenness of the field survey; and they may require lane closures for the inspected bridges and demand a lot of resources and time when used to survey an entire bridge deck or area (Oxems 2014).

As such, there is a need to develop more effective and efficient inspection techniques/alternatives that surpass the challenges associated with the conventional or state-of-the-art alternatives for modeling the deterioration of bridges. One such technique includes machine learning methods that are believed to overcome the issues associated with the different conventional alternatives as they possess many benefits including: accuracy, automation, speed, customizability, and scalability (Jain 2015); they allow time cycle reduction and efficient utilization of resources (Radhika and Latha 2019); their flexibility and capabilities in handling multidimensional and varied data in dynamic or uncertain environments (Gowri et al. 2019); few assumptions are needed to apply them (Codding 2013); they can be generalized over a large degree of data variation (Amin and Al-Darwish 2006); and their ability to discover patterns in data that exhibit significant unpredictable nonlinearity (Kanevski et al. 2004). In fact, the critical need for a data-driven approach for bridge deterioration based on data analytics was stressed by many previous studies. In relation to that, Liu and El-Gohary (2019) highlighted that: "bridge deterioration prediction is an indispensable component of bridge maintenance decision making. Recent studies have emphasized the need for taking a data-driven approach for better predicting bridge deterioration. The increasing availability of heterogeneous bridge data from different sources opens unprecedented opportunities to data analytics for better predicting deterioration and for learning how to better maintain bridges." In fact, equipping decision-makers with effective predictive tools is crucial in ensuring successful results (Assaad et al. 2020). That said, there is a clear need to provide data-driven methods that could be used in the evaluation and prediction of decks' conditions. To this end, this paper attempts to fill this critical need or knowledge gap through a comparative computational machine learning approach.

Since the conditions of bridges differ from one US state to the other, it is important to develop a BMS model for the prediction of deck conditions specific to each state. As such, triggered by their local environment, the poor conditions of bridges in Missouri, the funding issues associated with the MR&R of Missouri's bridges (Merkle and Myers 2006), and since there is no previous research effort that developed BMS for Missouri, the authors applied different machine learning algorithms to assess and predict the conditions of bridges in Missouri. However, it is worth noting that while this research is applied to bridges in Missouri, the technique is scalable and malleable enough to be used on any similarly available data set nationwide.

Conditions of Bridges in Missouri

As for any other structural systems, bridges become deficient or obsolete after they reach a certain critical age that reflects the necessity of repairing, maintaining, or replacing existing bridges (Koenigsfeld 2003). It is estimated that more than 40% of US bridges need to be repaired or replaced (Stone et al. 2002). Further, Missouri is one of many states that cannot accommodate the immediate repair and replacements needs of bridges (Merkle and Myers 2006). Missouri possesses the 7th largest system of roads and bridges in the US; however, it is ranked 46th in funding (Connolly 2018). Missouri has estimated a sum of $4.2 billion needed for the repair of around 4,800 bridges (ASCE 2018). The total number of bridges in Missouri was estimated to be 24,487 in 2017 with around 12.6% of them being structurally deficient (FHWA 2018a). This percentage is higher than the average US bridge deficiency rate of 9% (Kirk and Mallett 2018). In fact, Missouri is ranked 4th among US states in the total number of deficient bridges (ARTBA 2018). In addition, the average age of Missouri's bridges is 46 years with 60% of them exceeding the intended lifespan of 50 years (Connolly 2018). Furthermore, 1,194 of Missouri's bridges are considered to be weight restricted; meaning that they cannot carry some normal traffic (MoDOT 2018). Although congestion levels are low compared to other states, Missouri is considered to be in the bottom tier of US states in terms of the number of bad bridges (Miller 2015). The battle of repairing and replacing Missouri's bridges is becoming tougher with the increasing number of bridges that are assessed as "poor" each year (Connolly 2018). In addition, Missouri faces serious problems securing the necessary funds for the MR&R of its bridges. As such, there is a clear need to develop reliable and accurate BMS models that help transportation agencies in Missouri through their MR&R processes.

Methodology

The research methodology could be divided into three main steps: (1) data collection and data preprocessing, (2) data processing, and (3) model selection/evaluation as detailed in subsequent discussions. A summary of the methodological procedures is shown in Fig. 1. It is to be noted that Fig. 1 does not intend to show the exact sequence of steps since many steps were performed in parallel; but rather it shows the overall steps implemented in this paper. For example, data division was performed before data standardization.

Step 1: Data Collection and Preprocessing

Data Source and Data Cleansing

The National Bridge Inventory (NBI) is a database compiled by FHWA that publishes, in standardized format, data pertaining to all bridges in the US states. The data gathered over many years equips researchers with a valuable and reliable resource that helps predict the future conditions of bridges as well as in making informed decisions (Estes and Frangopol 2003). The data set used in the development of the BMS model in this paper is retrieved from FHWA's NBI database on all bridges in Missouri. According to FHWA, the deck condition reflects the overall in-place deterioration of the deck compared to the as-built condition. To have uniformity across bridge inspectors, FHWA's general condition ratings of decks are coded as presented in Table 1.

As can be seen from Table 1, the conditions of decks can be classified into categories. As such, the modeling of deck deterioration conditions is a classification problem rather than a regression framework. To this end, classification machine learning algorithms are developed and assessed. The deck conditions that were classified as N (not applicable) were dropped from the data set because this paper studies the deterioration of the decks of bridges and the N notation refers to bridges that do not possess decks as specified by FHWA (1995): "Code N for culverts and other structures without decks, e.g., filled arch bridge." In addition, the deck conditions rated as 0 (failed condition) or 1 (imminent failure condition) reflect that these bridges are "closed to traffic" and "out of service", respectively, as shown in Table 1. This means that the bridges are in a very severe condition and not in use, reflecting that the reconstruction of these bridges is required rather than inspections; thus they were dropped from the data set (there are a total of six bridges or an equivalent of 0.024% of all bridges in Missouri). A total data set of 19,269 bridges was considered in this paper. To this end, the developed machine learning classification algorithms will classify deck conditions into 8 categories: Condition 2 to Condition 9. Because the one hot encoding method substantially increases the dimensionality of the data, it is reasonable to treat the 8 categories of deck conditions as ordinal data since a higher condition number reflects a better deck condition and vice-versa. As such, the authors used the ordinal encoding method to make sure that the 8 deck categories are treated as ordinal data. The ordinal encoding method transforms the 8 categories to ordinal integers (0 to number of categories −1); readers can refer to Scikit-learn (2020) for details. The advantage of the ordinal encoding method is that it ensures that the categories are treated as ordinal, they are ordered, and that the order is maintained by the encoding (He and Parida 2016).

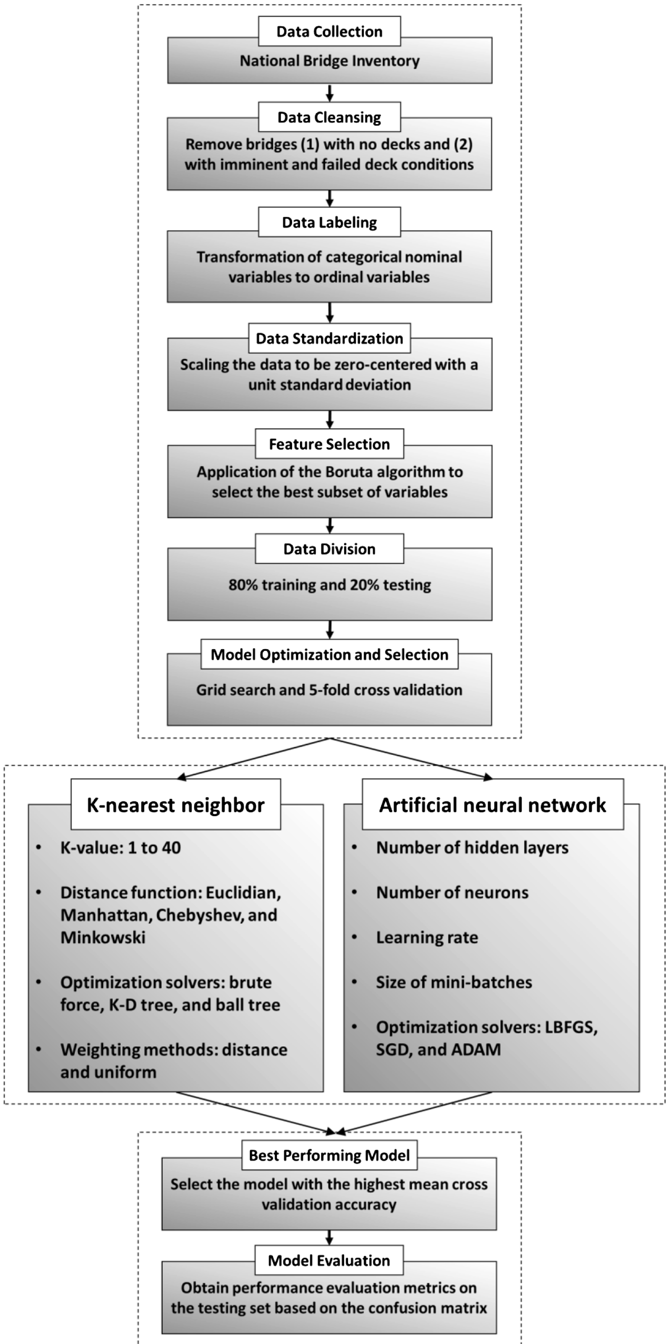

Fig. 1. Followed research methodology. The arrows do not reflect the exact sequence of the followed steps but rather show the overall steps implemented in this paper.

Coding and Software Packages

The data preprocessing and processing of the total 19,269 bridges were performed using Python which is an interpreted, high-level, general-purpose programming language. In addition, the classification machine learning algorithms were coded using Project Jupyter which is a nonprofit organization created to develop open-source software, open-standards, and services for interactive computing across different programming languages including Python. More specifically, Python's scikit-learn library was used to develop the classification machine learning algorithms; which is a free machine learning library. In addition, Python's open-source Pandas and Numpy libraries were used to add support for large, multidimensional arrays and matrixes, and data frames, along with a large collection of high-level mathematical functions to operate on the data.

Table 1. Deck conditions

Deck classification	Description
N	Not applicable.
Condition 9	Excellent condition.
Condition 8	Very good condition—no problems noted.
Condition 7	Good condition—some minor problems.
Condition 6	Satisfactory condition—structural elements show some minor deterioration.
Condition 5	Fair condition—all primary structural elements are sound but may have minor section loss, cracking, spalling, or scour.
Condition 4	Poor condition—advanced section loss, deterioration, spalling, or scour.
Condition 3	Serious condition—loss of section, deterioration, spalling, or scour have seriously affected primary structural components. Local failures are possible. Fatigue cracks in steel or shear cracks in concrete may be present.
Condition 2	Critical condition—advanced deterioration of primary structural elements. Fatigue cracks in steel or shear cracks in concrete may be present or scour may have removed substructure support. Unless closely monitored it may be necessary to close the bridge until corrective action is taken.
Condition 1	"Imminent" failure condition—major deterioration or section loss present in critical structural components or obvious vertical or horizontal movement affecting structure stability. Bridge is closed to traffic.
Condition 0	Failed condition—out of service—beyond corrective action.

Source: Data from FHWA (1995).

Data Preparation

The data that NBI publishes pertaining to bridges decks is presented in Table 2.

Since the variables/features present in Table 2 have varying magnitudes, units, and range (for instance the bridge age is in years while the structure length is in meters), this creates magnitude issues and thus there is a need to bring all features to the same level of magnitude (Asaithambi 2017). As such, data standardization was used to overcome this issue using Eq. (1). In addition, data standardization ensures that the distribution of all variables are zero-centered, and that the input features are standardized for efficient processing of the data (Vincent et al. 2019)

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where z = scaled value of the unscaled variable x that possesses a mean of μ and a standard deviation of σ.

It is worth mentioning that most machine learning algorithms do not handle categorical variables; and thus, categorical variables need to be transformed before being processed. Although some variables present in Table 2 are categorical variables (such as median type, service type, and construction type, among others), FHWA maps or codes each one of the possible values that the categorical variables can take to an ordinal/numerical scale representation. For instance, the FHWA's associated ordinal numerical codes or values for the categorical variable median are 0, 1, 2, and 3 representing no median, open median, closed median with no barriers, and closed median with nonmountable barriers, respectively. The same ordinal coding exists for the other categorical variables in Table 2.

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Table 2. Different variables and their description

Variable index	Variable	Description
1	Bypass, detour length	Indicates the actual length of the detour. The detour length should represent the total additional travel for a vehicle that would result from closing of the bridge
2	Bridge age	Represents the age of the bridge since its year of construction
3	Traffic lanes on the structure	Records and codes the number of lanes being carried by the structure
4	Average daily traffic	The average daily traffic volume for the bridge's inventory route
5	Approach road width	Represents the normal width of usable roadway approaching the structure
6	Median	Indicates if the bridge's median is nonexistent, open, or closed
7	Skew	Indicates the angle between the centerline of a pier and a line normal to the roadway centerline
8	Structure flared	Indicates if the structure is flared (i.e., the width of the structure varies). Generally, such variance will result from ramps converging with or diverging from the through lanes on the structure
9	Operations	Provides information about the actual operational status of the structure
10	Service type	The type of service on the bridge being highway, railroad, pedestrian-bicycle, etc.
11	Structure material	Indicates the kind of material for the main spans as concrete, steel, pre-stressed concrete, wood or timber, masonry, etc.
12	Construction type	Indicates the predominant type of construction as tee beam, box beam, truss deck, arch deck, etc.
13	Approach type	Indicates the type of approach spans to a major bridge or for the spans where the structure material is different
14	Approach construction	Indicates the predominant type of construction for approach spans
15	Number of spans	Records the number of spans in the main or major unit
16	Approach number	Records the number of spans in the approach spans
17	Horizontal clearance	Indicates the total horizontal clearance for the bridge's inventory route
18	Maximum span	Indicates the length of the maximum span measured along the centerline of the bridge
19	Structure length	Indicates the length of the structure; measured back to back of backwalls of abutments or from paving notch to paving notch
20	Left curb width	Represents the width of the left curb or sidewalk
21	Right curb width	Represents the width of the right curb or sidewalk
22	Roadway width	Indicates the most restrictive minimum distance between curbs or rails on the structure roadway.
23	Deck width	Indicates the out-to-out width of the deck
24	Vertical clearance	Indicates the actual minimum vertical clearance over the bridge roadway, including shoulders, to any superstructure restriction
25	Superstructure	Indicates the physical condition of all structural members of the superstructure of the bridge
26	Substructure	Indicates the physical condition of piers, abutments, piles, fenders, footings, etc.
27	Operating rating	Indicates the absolute maximum permissible load level to which the structure may be subjected
28	Inventory rating	Indicates the load level which can safely utilize an existing structure
29	Structure evaluation	Indicates the structure evaluation of the bridge in relation to the level of service which it provides
30	Deck geometry	Indicates the deck geometry evaluation in relation to the level of service which it provides
31	Posting evaluation	Evaluates the load capacity of a bridge in comparison to the State legal load
32	Alignment	Indicates the adequacy of the approach roadway alignment
33	Traffic direction	Indicates the direction of traffic of the bridge's inventory route
34	Deck material type	Records the type of deck system on the bridge as concrete cast in place, steel plate, open/closed grating, etc.
35	Wearing surface	Indicates the type of the wearing surface or protection system as monolithic concrete, integral concrete, epoxy overlay, bituminous, etc.
36	Membrane type	Indicates the type of the membrane as preformed fabric, built-up, etc.
37	Deck protection	Indicates the deck's protection as galvanized reinforcing, cathodic protection, internally sealed, etc.
38	Truck traffic	Indicates the average daily traffic associated to truck traffic

Source: Data from FHWA (1995).

Readers can refer to FHWA's (1995) report titled "*Recording and Coding Guide for the Structure Inventory and Appraisal of the Nation's Bridges*" for a complete understanding of how the categorical variables are represented as ordinal/numerical values. Since the one hot encoding method substantially increases the dimensionality of the data, the authors used FHWA's (1995) coding guidelines to transform the categorical variables to ordinal numerical values that were used for data processing for the developed machine learning algorithms.

Further, because of some incomplete/missing NBI data sets, data imputation was performed to ensure a sufficient and representative amount of data is processed into the machine learning algorithms (Camm et al. 2019). As such, multiple imputation (Ng et al. 2009) was performed since "the MI [multiple imputation] method was found to have superior performance to other techniques of handling missing data" (Wong et al. 2019), and it is perceived to be less susceptible to bias compared to other imputation approaches (Fraser et al. 2019). The multiple imputation is implemented by creating a set of imputations for the missing values in a stochastic way to allow for uncertainty about the missing data. This is accomplished by creating several different plausible imputed data sets and appropriately combining results obtained from each one of them (Sterne et al. 2009). The imputation method used in this paper is the multiple imputation with chained equations (MICE). MICE is considered one of the most advanced methodologies for performing missing data imputation (Drakos 2018). The parameters used for MICE are the generation of $m = 7$ imputed data sets with 30 iterations as recommended by van Buuren (2012), Harrell (2015), Moons et al. (2006), and White et al. (2011). The basic idea is to treat each variable with missing values as the dependent variable. Then, the MICE method cycles through all the different models and generates random draws from the predictive distributions determined by the fitted models. For more details on multiple imputation, readers can refer to Statsmodels (2020). The imputed data set was used for data processing.

Feature Selection

A subset of the features present in Table 2 are used in processing the data to include the variables/features that contribute the most to the predicted variable 'deck deterioration condition'. This is known as *feature selection* which is defined by Solorio-Fernández et al. (2020) as "the process of selecting the most useful features for building models in tasks like classification, regression or clustering. Moreover, feature selection not only reduces the dimensionality of the data facilitating their visualization and understanding; but also it commonly leads to more compact models with better generalization ability." In addition, the importance of feature selection lies in the fact that it "reduces overfitting by minimizing the redundancies in data, improves modeling accuracy, and decreases training time. It also reduces computational complexity and memory usage" (Ding et al. 2018).

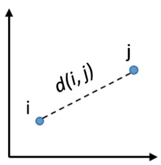
In general, different feature selection methods exist and they could be classified into two categories: filter and wrapper methods (Dy and Brodley 2004). The filter method relies on the statistical properties of the training data to select the features and does not incorporate any learning algorithm; whereas, the wrapper method relies on a learning algorithm and utilizes its performance for feature selection (Dy and Brodley 2004). Examples of filter methods include the correlation coefficient(s) or correlation matrix between the variables. Although some previous work has relied on such a method, it has "implicit orthogonality assumptions between variables, and the coefficient does not take mutual information between features into account. Additionally, this method only looks for linear correlations that might not capture many physical phenomena" (Mangal and Holm 2018). In other words, "correlation is highly deceptive as it does not capture strong non-linear relationships" (Asaithambi 2018). That said, improved wrapper feature selection methods have emerged in the recent years to address the drawbacks of the statistical filter methods. In fact, "the wrapper method is more effective than filter method since it considers the specific interactions between feature subset search and learning model" (Yan et al. 2019). To this end, the authors used the wrapper method to select the most relevant features. More specifically, the Boruta feature selection algorithm was used since it is "one of the best ways for implementing feature selection with wrapper methods" (Kaushik 2014). In addition, as provided by Cao et al. (2018), the Boruta algorithm: (1) facilitates the selection of the most relevant features with the greatest potential for enhancing prediction, (2) can select the most important features from a large number of variables even though the relationships among the features and the output variable are complex and nonlinear, and (3) is successful in finding the best feature subset to develop a forecasting model with outstanding accuracy. Bhalla (2017) summarizes the superiority of the Boruta feature selection algorithm by stating that: "it follows an all-relevant variable selection method in which it considers all features which are relevant to the outcome variable. Whereas, most of the other variable selection algorithms follow a minimal optimal method where they rely on a small subset of features which yields a minimal error on a chosen classifier." To implement the Boruta algorithm the following steps are followed: (1) create duplicate copies of all features; (2) the values in the created copies are permuted to remove any correlation between them and the output variable (the deterioration condition of decks), the created copies are called shadow variables; (3) combine the original variables with the shuffled copies; (4) run a random forest classifier on the expanded data set and perform a variable importance measure to evaluate the importance of each variable (higher values mean higher importance); (5) at every iteration, check whether a real feature variable has higher importance than the best of its shadow features and constantly remove feature variables that are perceived to be highly unimportant; and (6) repeat the previous steps until all features are classified as important or unimportant or until a specified iteration's limit is reached, whichever happens first.

Step 2: Data Processing

Choice of the Machine Learning Algorithms

Many machine learning algorithms are present in the literature; and it would be infeasible to examine all of them in-depth in a single research work. As such, the authors concentrated on reviewing the widely used algorithms that proved to yield good performance compared to other techniques. In relation to that, Ryu et al. (2019) and Jebelli et al. (2019) provided that the following machine learning algorithms are the most commonly used: *k*-nearest neighbors (KNN), multilayer artificial neural network (ANN), decision tree (DT), and support vector machine (SVM). Although all these machine learning algorithms were proven to possess good performance, each one of them has its own applications and drawbacks. In fact, the choice of the best algorithm(s) depends on the problem and the data in hand (Almaliki 2019). According to Veerashetty and Patil (2020) and Mohamed (2017), the main disadvantages of SVMs include: the inherent complexities resulting from the choice of the kernel function since making an incorrect selection can increase the percentage of error; they use quadratic programming that results in high algorithm complexity and extensive memory requirements; and their lack of transparency because they are considered a nonparametric method. On the other hand, according to Nayab (2020), many disadvantages are associated with DT including: instability (since the reliability of the information in the DT depends on feeding the precise internal and external information at the onset; modifications in the variables or alterations of the sequence can lead to major changes and might possibly require redrawing the tree), complexity (preparing DTs, especially large ones for complicated problems with many branches, is a complex and time-consuming task), limited capabilities (DTs examine only a single field at a time which leads to rectangular classification boxes; this may not correspond well with the actual distribution of records in the decision space), and the possibility of duplication with the same subtree on different paths. Given that this paper aims to develop a robust decision-support tool based on machine learning algorithms for the existing bridges in Missouri as well as for newly constructed bridges in Missouri, SVMs and DTs are not perceived to be the optimal choices for this paper.

On the other hand, the KNN machine learning algorithm performs well on large training sets (Topak et al. 2018; Richman 2011), and it is able to make new predictions on unseen data because of its ability to determine similarities among data (Miner et al. 2015). Furthermore, KNN is simple and easy to implement, does not require the tuning of several hyperparameters, does not make additional assumptions, and is versatile; it could be used for different applications (Horrison 2018). In fact, the KNN algorithm is considered one of the top 10 machine learning algorithms (Gou et al. 2019), and it is usually used as the baseline algorithm in many domain problems (Hu et al. 2016a). Furthermore, compared to other algorithms, KNN is able to generalize unseen data with potentially complex geometry (Chen et al. 2019). Similarly, ANNs are believed to be better than other machine learning algorithms (Ibrahim et al. 2019) as they are able to model complex problems since they possess the ability to control multidimensional data. In addition, they have the capability to achieve reliable prediction



$$d(i,j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2} \quad \text{Euclidean}$$

$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{ip} - x_{jp}| \quad \text{Manhattan}$$

$$d(i,j) = \lim_{h \to \infty} \left(\sum_{f=1}^{p} |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_{f}^{p} |x_{if} - x_{jf}| \quad \text{Chebyshev}$$

$$d(i,j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h} \quad \text{Minkowski}$$

Fig. 2. Investigated distances for the KNN machine learning algorithm.

performance by changing their structures and internal information during the training phase to model highly nonlinear systems. In fact, ANNs have proved their effectiveness in the evaluation and prediction of bridge deterioration as well as infrastructure asset management problems. For instance, Nguyen and Dinh (2019) developed an ANN model to predict the future conditions of the bridges in the state of Alabama using 2,572 bridges. Further, based on 942 data points, Huang (2010) developed an ANN model for bridge decks' deterioration in Wisconsin where the Pontis BMS is employed. In addition, Sirca and Adeli (2004) developed an ANN decision support system for estimating the detailed section properties of steel bridge girders needed in the load factor design-based rating. Assaad and El-adaway (2020b) developed an ANN for the assessment of hazard potentials of US dams. In addition, Yeung and Smith (2005) used ANN for detecting the damages in bridges by measurement of vibration signatures.

Based on the aforementioned information, the authors used ANN and KNN in this paper. It is worth mentioning that other algorithms might also be implemented; nevertheless, the work presented in this paper is believed to provide the foundation for, and encourage, future research to try other machine learning algorithms as it would not be reasonable to implement and compare all possible machine learning algorithms in a single study.

k-Nearest Neighbors

KNN is a machine learning algorithm that could be used for classification purposes. The output of the classification is determined as the class with the highest frequency from the *k*-most similar instances. In other words, each instance votes for its class and the class with the most votes is taken as the prediction (Brownlee 2019). The KNN algorithm "is one of the most widely used classification algorithms since it is simple and easy to implement. Moreover, it is usually used as the baseline classifier in many domain problems" (Hu et al. 2016a), and "because of the simplicity, effectiveness, and implementation of the KNN-based classification, KNN has been viewed as one of the top 10 algorithms in data mining" (Gou et al. 2019). Further, compared to other methods, KNN "allows an object recognition module to generalize to unseen objects with potentially complex geometry" (Chen et al. 2019). To this effect, and knowing that KNN was "asserted to be the most effective classifier for handling large sets" (Topak et al. 2018), it has been chosen to be implemented in this paper.

In the KNN machine learning algorithm, the inputs consist of the *k*-closest data points in the feature space. On the other hand, the output is a class membership (in this paper the deck deterioration condition). The condition of the deck is classified into one of the eight potential classes by a plurality vote of its neighbors, with the deck being assigned to the condition class that is the most common among its k neighbors (where k is positive integer that is data-dependent). Further, a KNN classifier "implements learning based on the nearest neighbors of each query point, where k is an integer value specified by the user . . . the optimal choice of the value k is highly data-dependent" (Scikit-learn 2019). As such, different values of k ranging from 1 to 40 were investigated in this paper, and the best value was determined using the grid search method.

To determine which of the k instances in the training set are most similar to the new input, a distance measure or function is used. It was demonstrated that "the chosen distance function can affect the classification accuracy of the *k*-NN classifier" (Hu et al. 2016a). In relation to that, there exist different distance measures including Euclidean, Manhattan, Chebyshev, and Minkowski (Scikit-learn 2019). Fig. 2 shows how these distances are calculated for two objects: $i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \ldots, x_{jp})$ described by p numeric attributes.

In addition to the *k*-value, these distance functions were considered the hyperparameters for the grid search method to optimize the KNN model's performance. Further, it was demonstrated that the choice of the KNN optimization solver affects the performance of the developed KNN algorithm (Mair et al. 2019). In relation to that, three different optimization solvers exist for solving KNN classification problems including: brute force, K-D tree, and ball tree (Scikit-learn 2019).

The brute force optimization algorithm is the most naïve neighbor search solver to compute the distances between the data points in the data set (Scikit-learn 2019). For N samples in D dimensions, the brute force solver scales as $O[DN^2]$. This algorithm has proved to be very competitive in small data samples since as N increases, the brute force approach might be computationally inefficient. Patel (2017) summarizes the brute-force algorithm by stating that it calculates the distance from a "point of interest (of whose class we need to determine) to all the points in training set. Then [the] class with majority points [is taken]. This is called brute force method."

The K-D tree is a tree-based optimization algorithm that attempts to reduce the required number of distance measures by efficiently encoding the aggregate distance information for the sample under investigation. It could be summarized as follows: "the basic idea is that if point A is very distant from point B, and point B is very close to point C, then we know that points A and C are very distant, without having to explicitly calculate their distance. In this way, the computational cost of a nearest neighbors search can be reduced to $O[DN \log(N)]$ or better" (Scikit-learn 2019). In addition, the K-D tree approach is very fast for low-dimensional neighbors searches; however, it might be computationally inefficient as D increases.

The ball tree optimization algorithm is a modified version of the K-D tree solver in the sense that K-D tree partitions data along Cartesian axes whereas ball tree partitions data in a series of nesting hyperspheres. This might require the construction of the tree more costly than other solvers, but it results in a data structure that is very efficient in high-dimensional data (Scikit-learn 2019). The ball tree partitions the data in a multidimensional space with a query time complexity of $O[\mathrm{D} \log(\mathrm{N})]$ (Patel 2017).

Table 3. KNN hyperparameters used in the grid search

Hyperparameter	Investigated values
k	1–40
Distance function	Euclidean, Manhattan, Chebyshev, and Minkowski
Weighting method	Uniform and distance
Optimization algorithm	Ball tree, K-D tree, brute force

As such, to ensure modeling accuracy and run-time performance (Mair et al. 2019), all three optimization solvers were investigated in this paper. In addition to the k-value and the distance functions, these three algorithms were considered to be the hyperparameters for the grid search method so as to optimize the KNN model performance.

Further, two weighting methods for the KNN algorithm exist: the uniform method and the distance method (Scikit-learn 2019). The uniform method equally weighs all points in each neighborhood whereas the distance method weighs the points by the inverse of their distance where closer neighbors have a greater influence than those that are farther away. These two weighting methods were considered in this paper. In relation to that and in addition to the k-value, the distance functions, and the optimization solvers, these weighting methods were considered to be the hyperparameters for the grid search method so as to optimize the KNN model performance. A summary of the hyperparameters considered in the grid search for the KNN machine learning algorithm is shown in Table 3.

Artificial Neural Networks

ANNs are "generally nonlinear mathematical learning algorithms that attempt to imitate some of the workings of the human brain" (Sarzaeim et al. 2017). They are considered "one of the most popular supervised learning techniques that can recognize the pattern between input and output data sets[, and] many researchers have been inspired to study and explore their computational capacities, specifically to identify and measure the extent of damage in structures" (Dawood et al. 2018). The structure of ANNs includes: individual parameters or weights for each node in the network's architecture, transfer/activation function that shapes the generated outputs or results, and learning rules that calculate the relative impact of the individual inputs (Mishra et al. 2017). ANNs are machine learning techniques that rely on adjusting individual parameters or weights by error minimization through learning from experience (Avci and Abdeljaber 2016).

The training of ANNs uses a supervised learning technique called backpropagation. The configuration or architecture of ANNs greatly affects their performance (Michael 2005; Keller et al. 2016). As such, it is crucial to study the effect of different hyperparameters that impact the prediction accuracy of ANNs. To this end, this paper investigates a multilayer perception classifier with different ANNs' configurations in terms of number of hidden layers, number of neurons, different learning rates, different sizes of mini-batches, and different computational solvers/optimizers. The architecture of ANNs is generally composed of: (1) an input layer that accepts input signals and redistributes them to all neurons in the hidden layers; (2) one or more hidden layers that include computing neurons for the processing of the input patterns; and (3) an output layer that accepts output signals from the hidden layers and reports the output pattern of the entire architecture (Michael 2005).

The number of neurons in the input layer is equal to the number of input variables/features, and the number of neurons in the output layer is equal to the number of output variables or classification classes (eight in this paper: Condition 2 to Condition 9). In this paper, the architecture of an ANN is represented as $(N1, N2, \ldots, Nn)$ where n is the number of hidden layers and N is the number of neurons in the corresponding hidden layer. The architecture of the ANN developed in this paper is shown in the "Results and Analysis" section. In general, the numbers of hidden layers and their neurons that provide the ultimate prediction/classification accuracy are determined through trial and error, and they are problem-dependent (Zhang et al. 2019). However, well-established guidelines exist to help researchers narrow their search space. That said, the authors used the guidelines provided by Heaton (2008) as shown in Table 4.

Table 4. Number of hidden layers and their descriptions

Number of hidden layers	Description
None	Only capable of representing linear separable functions or decisions.
1	Can approximate any function that contains a continuous mapping from one finite space to another.
2	Can represent an arbitrary decision boundary to arbitrary accuracy with rational activation functions and can approximate any smooth mapping to any accuracy.
>2	Additional layers can learn complex representations (sort of automatic feature engineering) for layer layers.

Source: Data from Heaton (2008).

Based on Table 4, the authors investigated 1, 2, 3, 4, and 5 hidden layers with different neurons within each one of them because they can represent simple separable problems, arbitrary decision boundaries, and complex representations. The number of hidden layers and the number of neurons (from 1 neuron to 10 neurons) in each hidden layer were considered as hyperparameters for the grid search method to optimize the ANN's performance.

Moreover, the rectified linear unit (ReLU) activation function $\mathrm{f}(x) = \max(0, x)$ was used since the output generated by the ANN could be any deck condition class belonging to Condition 2 to Condition 9. In addition, "The Relu function is the most used activation function in the neural network. It is a ramp function. It can increase the efficiency in the gradient descent and backward propagation. Generally, using Relu can avoid the problem of gradient explosion and gradient vanishing" (Lo et al. 2019). Further, using the ReLU function ensures that "the interdependence of parameters are reduced, which alleviates the occurrence of overfitting and reduces the training time of deep networks." (Li et al. 2019).

Further, it was demonstrated that the optimization algorithm affects the performance of the developed machine learning algorithm (Berahas and Takáč 2020). In relation to that, three different optimization algorithms exist for ANNs including: the limited-memory Broyden-Fletcher-Goldfarb-Shanno (LBFGS), the stochastic gradient descent (SGD), and the adaptive moment estimation (known as ADAM) (Scikit-learn 2019). In addition to the number of hidden layers and the number of neurons, these three algorithms were considered as the hyperparameters for the grid search method to optimize the ANN's performance.

The LBFGS solver is an optimization algorithm that belongs to the family of quasi-Newton methods that are considered a class of hill-climbing optimization techniques (Yilmaz and Oysal 2009). LBFGS solver approximates the Broyden-Fletcher-Goldfarb-Shanno algorithm, which is an iterative method for solving unconstrained nonlinear optimization problems, using a limited amount of computer memory and using an estimation to the inverse Hessian matrix (Elenchezhian et al. 2018). It is one of the commonly used algorithms for parameter estimation for machine learning classifiers, and is particularly well suited for optimization problems with a large number of variables because of its moderate memory requirement

Table 5. ANN hyperparameters used in the grid search

Hyperparameter	Investigated values
Number of hidden layers	1, 2, 3, 4, and 5
Number of neurons	1, 2, 3, 4, 5, 6, 7, 8, 9, and 10
Learning rate	0.1, 0.01, 0.001, and 0.0001
Size of mini-batches	16, 32, 64, 128, and 256
Optimization algorithm	LBFGS, SGD, and ADAM

(Liu et al. 2012). Limited by the length of the paper, readers who are interested in getting more details are advised to see Nocedal and Wright (2006).

SGD is an iterative optimization technique for solving an objective function that is differentiable or subdifferentiable. The gradient descent is one of the most popular algorithms to optimize neural networks (Ruder 2016). The algorithm is called stochastic since it utilizes randomly selected samples to evaluate the gradients. SGD performs a weight update for each training example xi and label yi. Limited by the length of the paper, readers who are interested in getting more details are advised to see Bottou (2012).

ADAM is an optimization algorithm that can be used in place of the classical SGD to update the weights in an iterative way based on the training data. It refers to a stochastic gradient-based optimizer (Scikit-learn 2019) proposed by Kingma and Ba (2014). The learning rate in ADAM is maintained for each weight and separately adapted as learning unfolds, whereas SGD maintains a single learning rate for all weight updates, and it does not change during training. ADAM works well on relatively large data sets in terms of both training time and validation score (Scikit-learn 2019). Limited by the length of the paper, readers who are interested in getting more details are advised to see Kingma and Ba (2014).

Furthermore, the learning rate and the size of mini-batches are considered to affect the performance of the ANN. As such, different learning rates (0.1, 0.01, 0.001, and 0.0001) and different size mini-batches (16, 32, 64, 128, and 256) where investigated in this paper as hyperparameters. In addition to the number of hidden layers, the number of neurons, the optimization algorithms, the different learning rates and size of mini-batches were considered hyperparameters for the grid search method as to optimize the ANN performance. A summary of the hyperparameters of the ANN machine learning algorithm is shown in Table 5.

Step 3: Data Division, Selection of Algorithm and Hyperparameters, and Model Evaluation

The information presented in this section aims to present all details pertaining to the data division, steps performed in relation to cross-validation, selection of the best performing machine learning algorithm and its hyperparameters, and model evaluation. Before training the machine learning models, the entire data set (19,269 bridges) was divided into 80% (15,415 bridges) as training/validation data and 20% (3,854 bridges) as testing data. The testing data (3,854 bridges) will be held out for final evaluation of the developed decision-support tool on unseen data to ensure its robustness. For the 80% training/validation data, fivefold cross validation was performed. All details pertaining to data division, selection of the best machine learning algorithm and its hyperparameters, and model evaluation are shown in Fig. 3.

The fivefold cross validation was conducted on the training/validation data set by splitting it into five folds such that the machine learning model is trained on four folds and its accuracy is reported/validated on the remaining single fold to select the best performing model and its optimal hyperparameters. This is repeated five times to ensure that the cross-validation set was run on the entire 80% training/validation data set. In simple terms, the authors used the fivefold cross validation on the training/validation where the 15,415 bridges were divided into five equal folds with the first fold (3,083 bridges) being used as a cross-validation set and the four other folds (12,332 bridges) to train the machine learning algorithms. In this first iteration, the accuracy on the cross-validation set (CV1) is reported. In the second iteration, the second fold is used as a cross-validation set (3,083 new bridges other than the previously used 3,083 bridges) and the four other folds (12,332 bridges) are used to train the machine learning algorithms. In this second iteration, the accuracy on the cross-validation set (CV2) is reported. This is repeated five times as to ensure that the cross-validation set was run on each one of the five folds. The results of the fivefold cross validation methods are five accuracies (CV1, CV2, CV3, CV4, and CV5) as shown in Fig. 3.

Once the cross-validation process has been repeated five times, the average accuracy over the five rounds was calculated as shown in Fig. 3, and it is used as the basis for the optimal model and hyperparameters' selection (Hammerla and Plötz 2015). It is worth mentioning that the exhaustive grid search method was used to

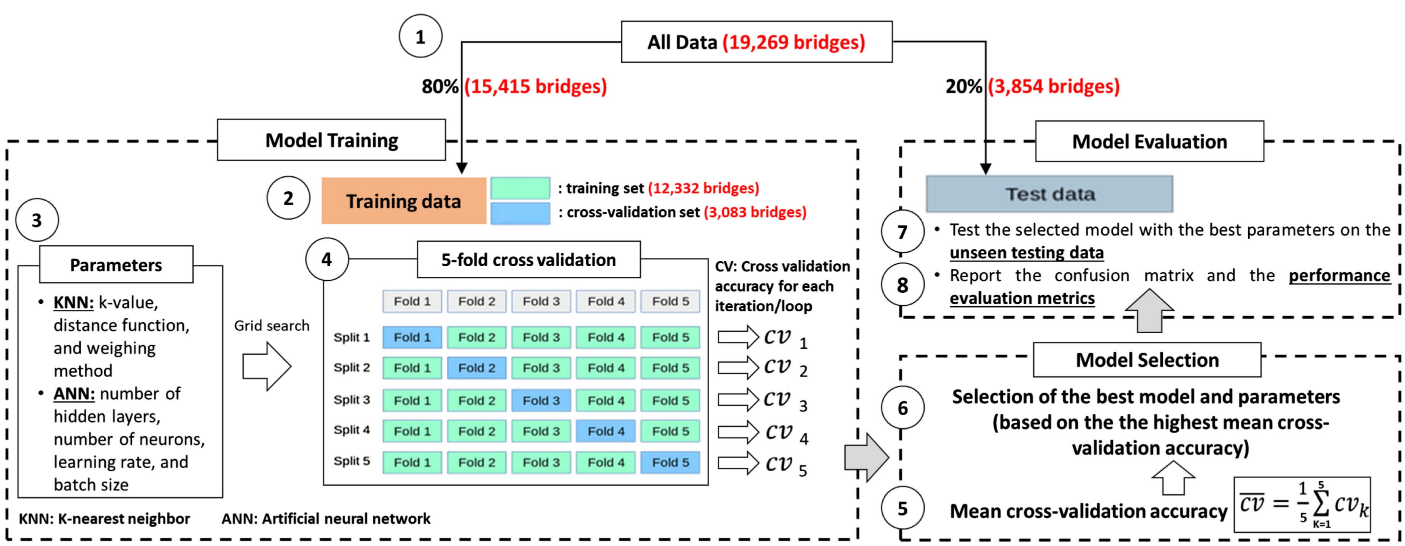


Fig. 3. Steps followed with regards to data division, model training, model selection, and model evaluation.

loop over all possible hyperparameters' values for the investigated machine learning algorithms (both KNN and ANN) since it is the most commonly used approach (Chowdhury et al. 2019). Grid search is the process of building a model on each hyperparameter combination and iterating through each combination accordingly to select the optimal hyperparameters of the given model (Zhao and Jiang 2019). Further, the early stopping regularization was used to prevent the machine learning algorithms from being overfitted or underfitted (Taormina and Galelli 2018). The early stopping ensures that the training of the machine learning algorithm is terminated when there is no improvement in the accuracy.

Once the best model and its optimal hyperparameters are selected based on the highest obtained mean fivefold cross validation accuracy, the selected model is evaluated on the unseen testing data set (3,854 bridges) for final testing of the created decision-support tool or machine learning model. In relation to that, the confusion matrix is obtained and the different performance evaluation metrics are calculated including accuracy, recall, and precision as provided in Eqs. (2)–(4), respectively. It is worth mentioning that the accuracy is calculated for the overall machine learning model whereas the recall and precision are obtained for each individual bridge deck condition. The used data division and fivefold cross-validation protocol in this paper is considered to be better than other methods as it ensures the generalization and robustness of the created model (Hong et al. 2018), and it avoids overfitting in the predictive machine learning algorithms (Chemchem et al. 2019)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

where TP = true positives; TN = true negatives; FP = false positives; and FN = false negatives.

Results and Analysis

The Boruta algorithm was applied for feature selection as detailed in the "Methodology" section, and the obtained results are shown in Fig. 4. As it can be seen from Fig. 4(a), at Iteration 0, all the features/variables present in Table 2 were classified as tentative; meaning they are waiting to be classified as important/relevant to the deck condition or unimportant/irrelevant to the deck condition. When the number of iterations progress, these variables start to be classified as important and unimportant as shown in Fig. 4(a). This continues until no single variable is classified as tentative as shown in Fig. 4(a) (the Boruta algorithm stopped after 322 iterations when the number of tentative variables is zero). That is, at Iteration 322, all variables were classified as being either important or unimportant. That said, 10 variables/features were selected to be the best subset that is relevant to the deck condition. These 10 variables are shown in Fig. 4(b). Table 6 shows a statistical summary for the response and the obtained predictive variables.

The KNN machine learning algorithm was applied to predict the deck conditions of bridges. As provided in the "Methodology" section, the developed KNN model was optimized using the grid search with different k values, optimization solvers, distance functions, and weighting methods. Since the total number of all possible combinations of these different hyperparameters is huge and equal to 960 $[(40 \text{ values for } k) \times (3 \text{ solvers}) \times (4 \text{ distance functions}) \times (2 \text{ weighting methods})]$, it is not feasible to report all individual accuracies for each possible combination obtained using the grid search method. As such, the authors report the obtained highest accuracy and the associated best combination of hyperparameters as provided in Table 7. As shown in Table 7, the best KNN model corresponds to a k value of 22, a Manhattan distance function, the distance weighting method, and a K-D tree solver with a mean cross-validation accuracy of 89.88%.

The ANN machine learning algorithm was trained to predict the deck conditions of bridges. By the same token, since the total number of all possible combinations of the different hyperparameters is huge, it is not feasible to report all individual accuracies for each possible combination obtained using the grid search method. As such, the authors report the highest accuracy obtained and the associated best combination of hyperparameter as provided in Table 8. As shown in Table 8, the best ANN model corresponds to 4 hidden layers with 10 neurons in each hidden layer, a learning rate of 0.01, a batch size of 256, and an ADAM solver with a mean cross-validation accuracy of 91.76%.

Based on the previously obtained results for KNN and ANN as shown in Tables 7 and 8, respectively, the model with the best performance is the one corresponding to the ANN machine learning algorithm with the simple architecture depicted in Fig. 5.

To validate and evaluate the performance of the selected best model shown in Fig. 5, the different performance evaluation metrics were obtained based on the confusion matrix for the 20% unseen testing data set as detailed in the "Methodology" section. The obtained confusion/classification matrix as well as the performance evaluation metrics (precision, recall, and accuracy) of the model are shown in Fig. 6.

As can be seen from Fig. 6, the highest numbers exist on the diagonals of the confusion matrix where the rows represent the true deck condition and the columns represent the predicted deck condition. This reflects that most of the predictions are correct. In fact, the total accuracy of the model is equal to 91.44% on the testing set as shown in Fig. 6. In addition, Fig. 6 shows the obtained recall and precision for the individual deck conditions. A summary of the characteristics and performance of the best machine learning model is shown in Table 9. As shown in Table 9, the prediction accuracy on the training test was equal to 91.92%. Since the testing and training accuracies are close, no overfitting happened. This is also reflected from the fact that the obtained accuracy of 91.44% in Fig. 6 is very close to the mean cross-validation accuracy of 91.76% (Table 8).

The obtained accuracy of 91.44% is considered to be an acceptable accuracy when compared to other studies that tried to predict the deck condition of bridges in different US states. As an example, the developed ANN model by Nguyen and Dinh (2019) to predict the deck condition rating of the bridges in Alabama resulted in an accuracy of 76.6%. In addition, the developed ANN model by Liu and El-Gohary (2019) to predict the deck condition of the bridges in nine states: Illinois, Michigan, Delaware, Idaho, Arkansas, South Carolina, Utah, Nevada, and Wyoming resulted in an accuracy of 89.7%. To this end, the developed machine learning algorithm in this paper could be used as a simple, yet accurate, method or decision-support tool to evaluate, assess, and predict the deck deterioration condition by transportation agencies in Missouri.

Interpretations and Discussions

Since most of the machine learning algorithms lack interpretability, this section aims to interpret and discuss the previously reported

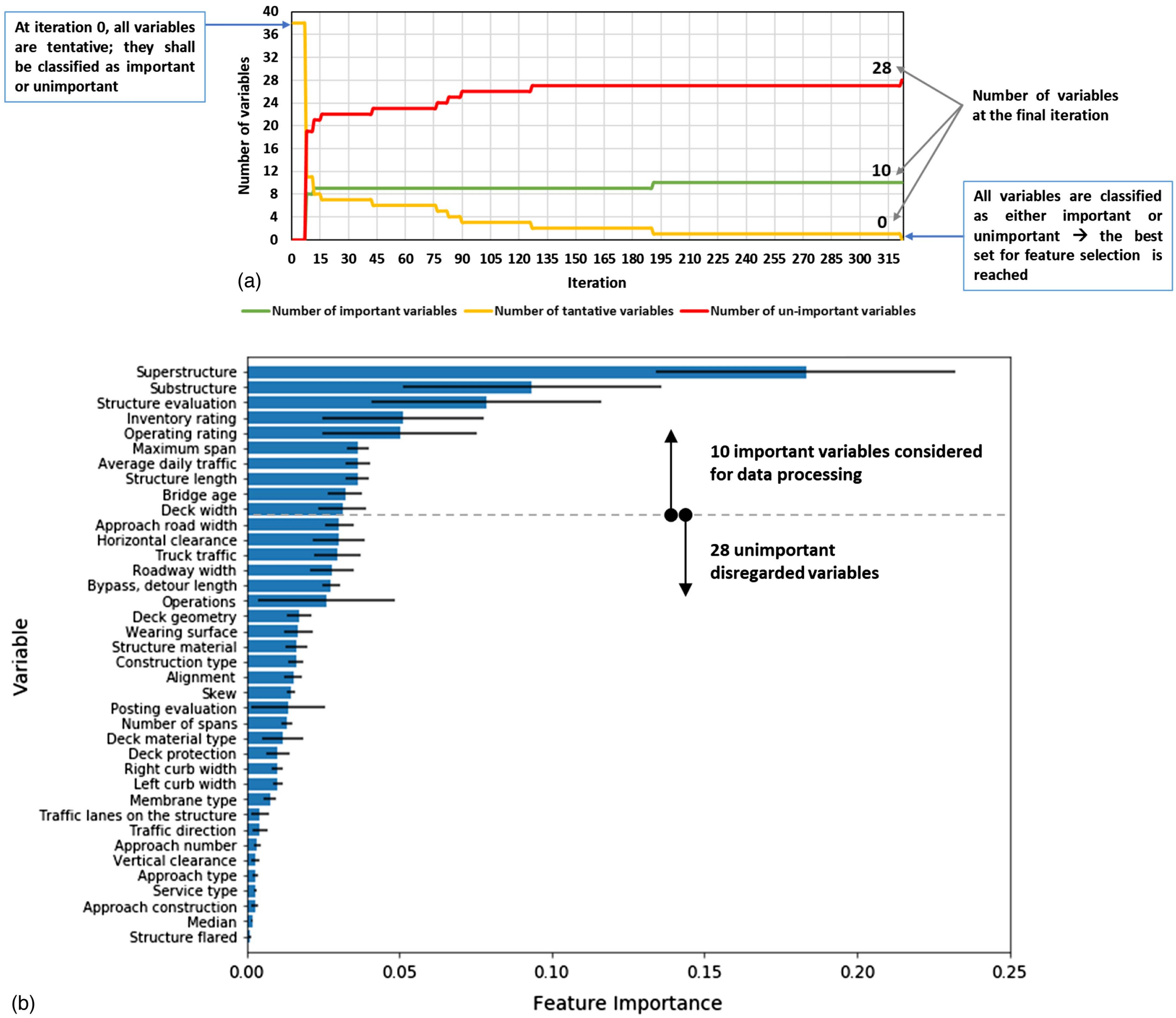


Fig. 4. Results of the feature selection.

Table 6. Statistical summary of response and predictive variables

Variables	Count	Mean	Standard deviation	Min	25%	50%	75%	Max
Deck condition	19,269	6.79	1.27	2.0	6.0	7.0	8.0	9.0
Bridge age	19,269	41.47	29.95	1.0	18.0	31.0	59.0	145.0
Average daily traffic	19,269	3,552.36	11,384.01	1.0	30.0	100.0	1,531.0	198,800.0
Maximum span	19,269	17.37	13.48	0.8	9.9	15.1	21.6	457.2
Structure length	19,269	45.40	87.21	6.1	12.8	25.9	52.4	2,391.8
Deck width	19,269	8.88	4.91	3.0	6.2	7.5	9.8	104.5
Superstructure	19,269	6.80	1.26	0.0	6.0	7.0	8.0	9.0
Substructure	19,269	6.78	1.38	0.0	6.0	7.0	8.0	9.0
Operating rating	19,269	52.14	23.43	0.0	34.2	53.1	68.4	99.9
Inventory rating	19,269	30.78	15.85	0.0	18.0	31.5	39.6	99.9
Structural evaluation	19,269	5.81	1.86	0.0	5.0	6.0	7.0	10.0

Table 7. Obtained highest accuracy for the KNN algorithm and the associated hyperparameters

K value	Distance function	Weighting function	Solver	Mean cross-validation accuracy (%)	± Standard deviation (%)
22	Manhattan	Distance	K-D tree	89.88	1.51

Table 8. Obtained highest accuracy for the ANN algorithm and the associated hypersparameters

Hidden layers	Number of neurons in each hidden layer	Learning rate	Batch size	Solver	Mean cross-validation accuracy (%)	± Standard deviation (%)
4	10	0.01	256	ADAM	91.76	1.48

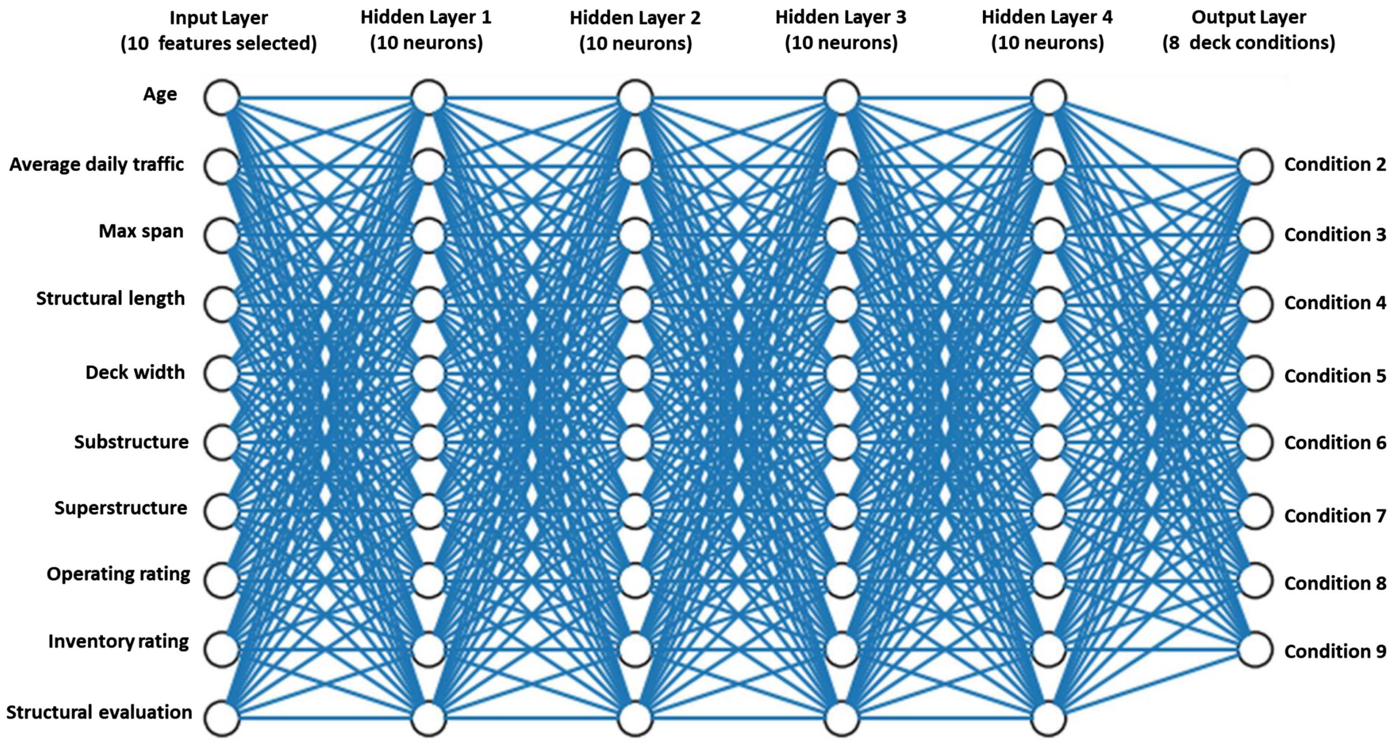


Fig. 5. Selected best machine learning algorithm.

True Deck Condition	Predicted Deck Condition: Condition 2	Condition 3	Condition 4	Condition 5	Condition 6	Condition 7	Condition 8	Condition 9	Recall (%)
Condition 2	1	0	0	0	0	0	0	0	100
Condition 3	0	70	1	1	0	0	0	0	97.22
Condition 4	0	2	120	1	1	0	0	0	96.77
Condition 5	0	2	6	343	16	2	0	0	92.95
Condition 6	0	1	3	16	725	4	2	0	96.54
Condition 7	0	0	0	23	65	1321	59	17	88.96
Condition 8	0	0	0	5	7	70	685	2	89.08
Condition 9	0	0	0	0	2	8	14	259	91.52
Precision (%)	100	93.33	92.31	88.17	88.85	94.02	90.13	93.17	91.44 Accuracy

Fig. 6. Confusion matrix and performance evaluation metrics.

Table 9. Best performing machine leaning model for bridge deck conditions in Missouri

Machine learning algorithm	Number of hidden layers and neurons	Optimization solver	Learning rate	Batch size	Activation function	Training accuracy (%)	Testing accuracy (%)
ANN	(10, 10, 10, 10)	ADAM	0.01	256	Relu	91.92	91.44

results. As seen from Fig. 4(b), the first two most important features are the superstructure and substructure of the bridge, respectively. As shown in Fig. 7, the superstructures are the structural systems that support the decks of bridges and the substructure includes the piers, abutments, and foundations of the bridge. Therefore, it is rational to anticipate that the superstructure and the substructure would have the highest impact on the conditions of decks.

According to Fig. 4(b), the other features classified as important include: operating rating, deck width, average daily traffic, structural evaluation, inventory rating, structure length, the bridge age,

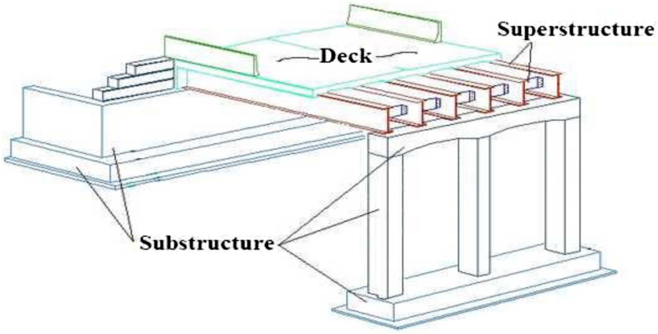


Fig. 7. Structural elements of a typical highway bridge. (Adapted from MDoT 2018.)

and maximum span. These findings were in line with previous research studies that investigated bridge deterioration (Carmichael et al. 2014; Cheng and Melhem 2005; Liu and El-Gohary 2019; Nguyen and Dinh 2019). To this end, the 10 variables obtained by the feature selection [shown in Fig. 4(b)] are considered to be the most important variables that impact the condition of bridge decks in Missouri. That said, Missouri's transportation agencies shall closely monitor and manage this set of variables to minimize the deterioration of their current bridges if possible, or ultimately, the deterioration of future bridges.

It is worth mentioning that false predictions, which are reflected by the precision, recall, and accuracy evaluation metrics, are inevitable in any developed decision-support tool or machine learning algorithm as it is impossible to get 100% perfect predictions for the output variable; especially for highly complex decisions and modeling problems such as bridge decks' deterioration conditions. That said, it is not possible to completely avoid false predictions; however, it is possible to minimize them or deal with them; especially those related to dangerous deck conditions. This could be incorporated into the modeling approach followed by the transportation and bridge authorities by setting a weighted importance for the predicted conditions. In other words, transportation and bridge agencies can impose a high-cost penalty on the mispredicted decks with dangerous conditions as compared to the costs or weights of the bridge decks with better or less dangerous conditions. In simple terms, bridge agencies would penalize the provided decision by the developed machine learning model for bridge decks predicted to have dangerous conditions as compared to other less critical deck conditions. Another possible approach to deal with or minimize the error of the falsely predicted bridge decks with dangerous conditions is the reporting of the confidence interval for the predicted conditions by the authorities responsible for the management of bridges. Relying on the obtained confidence interval, bridge authorities can then take their final decision based on the predicted deck condition. In relation to that, the radius r of the confidence interval would be obtained using Eq. (5) which is known as the Wilson score interval. In relation to Eq. (5), bridge authorities would increase the significance level to 98% or 99% before reaching a conclusive decision on the condition of the deck being dangerous.

$$r = z \times \sqrt{\frac{\text{accuracy} \times (1 - \text{accuracy})}{n}} \tag{5}$$

where r = radius of the confidence interval, z = desired confidence level ($z = 1.64$ for 90% confidence, $z = 1.96$ for 95% confidence, $z = 2.33$ for 98% confidence, and $z = 2.58$ for 99% confidence), and n is the number of bridges.

Practical Contributions

Compared to other structural elements, bridge decks are more susceptible to severe deterioration because they are exposed to harsh conditions such as heavy traffic, varying temperatures, road salts, and abrasive forces (Alberta Transportation 2003). In fact, the inspection of bridges is not simple and is performed to identify different defects in the decks including: cracking, scaling, spalling, leaching, chloride contamination, potholing; among others. State departments of transportation and the USDOT have been facing escalating challenges for decades because of the deterioration of bridge decks. In addition, the current and future conditions of bridge decks are considered to affect the amount of the future bridge funding needs and adversely impact the serviceability of bridges.

Since the visual formal inspection requires huge effort, much time, high costs, and even substantial danger (Gillins et al. 2016), the available bridge data opens unprecedented opportunities for machine learning algorithms and data analytics for better evaluation and prediction of bridge deck conditions. Despite the fact that machine learning models are referred to as black-boxes, the evaluation of the conditions of bridge decks using the developed computational framework has many practical contributions including:

- Its implementation as an easy to use and adopt decision-support tool by transportation agencies and bridge authorities to assess, evaluate, and predict the conditions of their bridge decks with good accuracy.
- The use of the proposed model could provide improved access to a large amount of information on bridge deterioration and maintenance actions, which have typically been unexploited and buried in bridge inspection reports (Liu and El-Gohary 2017).
- The extracted information based on the developed machine learning model for the deterioration of bridge decks and the factors that most impact their conditions could create new knowledge on existing bridge deck deficiencies and maintenance strategies, improve the understanding of the deterioration of bridge decks, and lead to enhanced bridge deck deterioration predictions and better maintenance decision making.
- The developed decision-support tool possesses many benefits compared to the formal visual inspection of bridge decks including: accuracy, automation, speed, customizability, scalability, time-cycle reduction, efficient utilization of resources, ease of implementation, and no requirement of substantial investments.
- The developed model can be generalized over a large degree of bridge data variations in the state of Missouri because the model was evaluated on unseen deck conditions and reported good prediction accuracy. This reflects its robustness in making generalized predictions to any new bridge decks that were not used in the training of the algorithm.
- The use of the developed model for assessing and predicting deck conditions does not create disruption to the public nor does it necessitate lane closures for the inspected bridge decks, or any interruptions to the traffic and commuters.
- The proposed machine learning framework could be used as a predictive maintenance tool and thus help in setting optimized maintenance strategies in terms of determining the best maintenance schedule, prioritization of maintenance needs, and elimination of defects in decks at an early stage.
- The developed decision tool in this paper is believed to be unique as there is no similar framework present in the literature

for the state of Missouri (which possesses the 7th largest system of roads and bridges in the US).

- While this research is applied to bridges in Missouri, the research methodology is scalable and malleable to be used on any similar available data set nationwide.
- It could help in addressing the lack of resources and the funding issues associated with the MR&R of Missouri's bridges (Merkle and Myers 2006) as better allocation and distribution of funds could be entertained based on the predicted deck conditions using the developed machine learning framework.
- It is considered to be a more effective and efficient alternative to conventional methods as it surpasses the challenges associated with many state-of-the-art techniques used for modeling, assessing, and forecasting the deterioration of bridge decks.

The practical contributions of this research work are also reflected by the critical issues that are facing US bridges, which crystalizes the need for alternative methods to help minimize or avoid the devastating consequences of bridge deterioration inefficiencies. This is reflected by the fact that many US states, including Missouri, possess very tight budgets allocated for the continuous planning of the MR&R of their bridge decks. In fact, the available budgets are very far from covering the costs of inspection or any needed maintenance in many states, like Missouri, that cannot accommodate the immediate repair and replacements needs of its bridges (Merkle and Myers 2006). This has caused Missouri and many other states to be behind in bridge deck inspections where many bridges go longer than several years without inspection. In relation to that, the efforts to monitor and repair bridge decks in Missouri are piecemeal and drastically underfunded, and hundreds of bridge decks whose failure would put lives in danger are years overdue for inspection. This is reflected by the fact that around 1,194 of Missouri's bridges are considered to be weight restricted; meaning that they cannot carry some normal traffic (MoDOT 2018). Moreover, Missouri is considered to be among the bottom tier of US states in terms of the number of bridges needing attention (Miller 2015), and the battle of repairing and replacing Missouri's bridges is becoming more and more demanding and challenging (Connolly 2018).

Based on all aforementioned information, the proposed machine learning algorithm has many potential benefits and substantial practical opportunities and contributions. In fact, recent studies have emphasized the need for taking a data-driven approach for better predicting bridge deterioration conditions. This is triggered by the increasing availability of heterogeneous bridge data from different sources; which opens unprecedented opportunities for data analytics to better predict bridge deteriorations and to learn how to better maintain bridges (Liu and El-Gohary 2019).

Conclusion and Future Work

This research attempted to develop a BMS to evaluate and predict bridge deck deterioration conditions in Missouri. In relation to that, this paper (1) identified the best subset of 10 variables that affect the classification condition of bridge decks in Missouri; (2) investigated the performance of two machine learning techniques: ANN and KNN; and (3) optimized the accuracy of the developed classification machine learning algorithms through finding the best set of hyper-parameters that lead to the highest performance. To this end, and after developing two machine learning models (ANN and KNN) and optimizing/tuning their hyperparameters, the ANN algorithm with the (10, 10, 10, 10) architecture, the ADAM optimization algorithm, a learning rate of 0.01, and a batch size of 256 yielded better accuracy compared to the KNN algorithm. This study adds to the body of knowledge by devising a computational data-driven framework that is valuable for transportation agencies as it allows them to evaluate and predict deck conditions with good accuracy. Although the developed BMS model predicts the deck condition with high accuracy, it is not intended to provide detailed reflection of the structural conditions of bridge decks. However, it is advisable to use the developed BMS model in conjunction with other technology-based methods such as: ultrasonic surface wave techniques, ground penetrating radar, sensors, etc. Because no costs are associated with the use of the developed BMS model and since considerable time savings are entertained as well, it is recommended that transportation agencies utilize the developed BMS framework as a first step for the prediction of deck conditions; supported with any other available decision information, if any. The developed model also provided insights on the factors affecting the conditions of bridge decks. This empowers transportation agencies to direct their efforts toward ameliorating these critical factors for current bridges if possible, or ultimately for future bridges. As such, this will ensure proper and more effective distribution of funds allocated for the MR&R of bridges.

Since data is easily accessible, future work could be to devise a similar BMS model for other states. This will result in having a set of reliable BMS frameworks that are able to predict the deck condition of bridges specific to each state. Other possible future work is the development of a more holistic and comprehensive BMS model for the prediction of deck conditions for mid-to-long bridge spans as well as different bridge components that even go beyond decks (i.e., substructure and superstructure). This process could be facilitated by incorporating technology-based techniques as well. This should result in factoring a comprehensible list of variables affecting overall bridge conditions. Also, from a methodological point of view, this research could be extended and altered as appropriate toward developing a value-engineering based approach for evaluating the potential usage of different materials and methods for different infrastructure applications like pavements, geotechnical considerations, and others.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

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